

Satoru Kobayashi **Assistant Professor**

Departmental Address: Department of Biomedical Sciences, NYIT College of Osteopathic Medicine, Northern Boulevard, Old Westbury, New York 11568

Education and postgraduate training: (list in descending chronological order)

Aug. 23 2004-Sep. 30 2012 Postdoctoral Fellow, Cardiovascular Health Research Center (formerly Cardiovascular Research Institute), Sanford Research /USD, SD 57104 United States
Jan. 23 2003-Jun. 30 2004 Postdoctoral Fellow, Department of Biology, University of South Dakota, SD 57069 United States
Oct. 1 2002-Dec. 31 2002 Postdoctoral Researcher, Cryobiosystem Research Institute, Iwate University, Japan
Apr. 1 1999-Aug.31 2002 PhD in The United Graduate School of Agricultural Sciences, Iwate University, Iwate Japan
Apr. 1 1997-Mar. 31 1999 MS in Faculty of Agriculture, Iwate University, Iwate Japan
Apr. 1 1993-Mar. 31 1997 BS in Faculty of Agriculture, Iwate University, Iwate Japan

Clinical, Academic, and Administrative Appointments: (list current and former academic appointments and years)

Sep. 1 2022-Present Assistant Professor, New York Institute of Technology College of Osteopathic Medicine (NYITCOM) Department of Biomedical Sciences, NY 11568 United States
Jul. 29 2013-Aug. 31 2022 Instructor, NYITCOM Department of Biomedical Sciences, NY 11568 United States
Oct. 1 2012-Jun. 30 2013 Staff Scientist, Cardiovascular Health Research Center, Sanford Research /USD, SD 57104 United States
Oct.1 1999-Mar. 31 2002 Research Assistant, Department of Agriculture, Iwate University, Japan

Peer reviewed papers (list in descending chronological order) (these may include “accepted” and “in press” manuscripts)

1. Chen K, Timm D, Huang Y, **Kobayashi S**, Liang Q. Caloric Restriction Attenuates Oxidative Injury but Fails to Reverse Established Cardiac Remodeling in Atrial Natriuretic Peptide Receptor 1-Deficient Mice. International Journal of Drug Discovery and Pharmacology (IJDDP). 2026, 5(2): 100015. doi: 10.53941/ijddp.2026.100015
2. Khinchin J, Rakoubian A, Romano V, Ryan T, Yarbrow J, **Kobayashi S**, Liang Q. Repurposing metformin for cardioprotection: mechanisms and therapeutic potential across cardiovascular pathologies. Front Pharmacol. 2026 Feb 2;17:1681783. doi: 10.3389/fphar.2026.1681783. PMID: 41704279; PMCID: PMC12907551.

3. Rakoubian A, Khinchin J, Yarbrow J, **Kobayashi S**, Liang Q. Isoform-Specific Roles of AMP-Activated Protein Kinase (AMPK) in Cardiac Physiology and Pathophysiology. *Front Cardiovasc Med*. 2025;12. doi:10.3389/fcvm.2025.1638515
4. Chong L, Dushaj N, Rakoubian A, Yarbrow J, **Kobayashi S**, Liang Q. Unraveling the Roles of HIF-1, HO-1, GLUT-1 and GLUT-4 in Myocardial Protection. *International Journal of Drug Discovery and Pharmacology (IJDDP)*. 2024 August; 3(3).
5. Nakayama Y, **Kobayashi S**, Masihuddin A, Abdali SA, Seneviratne AMPB, Ishii S, Iida J, Liang Q, Yoshioka J. Systemic Deletion of ARRC4 Improves Cardiac Reserve and Exercise Capacity in Diabetes. *Circ Res*. 2024 Jul 19;135(3):416-433. doi: 10.1161/CIRCRESAHA.123.323158. Epub 2024 Jul 1. PubMed PMID: 38946541; PubMed Central PMCID: PMC11257811.
6. Van J, Hahn Y, Silverstein B, Li C, Cai F, Wei J, Katiki L, Mehta P, Livatova K, DelPozzo J, Kobayashi T, Huang Y, **Kobayashi S**, Liang Q. Metformin Inhibits Autophagy, Mitophagy and Antagonizes Doxorubicin-Induced Cardiomyocyte Death. *Int J Drug Discov Pharm*. 2023 Mar;2(1):37-51. doi: 10.53941/ijddp.0201004. Epub 2023 Feb 17. PubMed PMID: 38487671; PubMed Central PMCID: PMC10939033. (Selected the Best Article 2024)
7. **Kobayashi S**, Hahn Y, Silverstein B, Singh M, Fleitz A, Van J, Chen H, Liang Q. Lysosomal dysfunction in diabetic cardiomyopathy. *Front Aging*. 2023 Jan 20;4:1113200. doi: 10.3389/fragi.2023.1113200. PMID: 36742461; PMCID: PMC9894896.
8. Greco LV, Migirov A, Ojamaa K, Li Y, Huang Y, **Kobayashi S**, Udo-Bellner L, Stout R, Cohen TJ, Zhang Y. Stabilizing Cardiac Ryanodine Receptor with Dantrolene Treatment Prevents Binge Alcohol Enhanced Atrial Fibrillation in Rats. *J Cardiovasc Pharmacol*. (2022) Aug 9. doi: 10.1097/FJC.0000000000001346. Epub ahead of print. PMID: 35947104.
9. Singh M, Nicol AT, Del Pozzo J, Wei J, Singh M, **Kobayashi S**, Liang Q. Demystifying the relationship between Metformin, AMPK and Doxorubicin Cardiotoxicity. *Frontiers in Cardiovascular Medicine*. (2022) Doi:10.3389/fcvm.2022.839644
10. Klionsky DJ, Abdel-Aziz AK, Abdelfatah S, **Kobayashi S** et al. Guidelines for the use and interpretation of assays for monitoring autophagy (4th edition)1. *Autophagy*. 2021;17(1):1-382. Doi:10.1080/15548627.2020.1797280
11. **Kobayashi S**, Zhao F, Zhang Z, Kobayashi T, Huang Y, Shi B, Wu W, Liang Q. Mitochondrial Fission and Mitophagy Coordinately Restrict High Glucose Toxicity in Cardiomyocytes. *Front Physiol*. 2020 Dec 10;11:604069. Doi: 10.3389/fphys.2020.604069. PMID: 33362579; PMCID: PMC7758327.

12. Zhao F, **Kobayashi S**, Zhao F, Li C, Cai F, Huang Y, Liang Q, Hui S. Comprehensive Lipidomic Profiling in Serum and Multiple Tissues from a Mouse Model of Diabetes. *Metabolomics*. Accepted on Sep 24 2020. Doi: 10.1007/s11306-020-01732-9. PMID: 33067714
13. **Kobayashi S**, Patel J, Zhao F, Huang Y, Kobayashi T, Liang Q. Novel Dual-Fluorescent Mitophagy Reporter Reveals a Reduced Mitophagy Flux in Type 1 Diabetic Mouse Heart. *J Am Osteopath Assoc*. 2020, 120 (7), 446–455. Doi:10.7556/jaoa.2020.072. PMID: 32598458
14. **Kobayashi S**, Zhao F, Kobayashi T, Hagiwara M, Kaminaris A, Li C, Cai F, Huang Y, Liang Q. Hyperglycemia-induced cardiomyocyte death is mediated by lysosomal membrane injury and aberrant expression of cathepsin D. *Biochemical and Biophysical Research Communications*. 2020 Feb 26;523(1):239-245, Published Online:18 Dec 2019. Doi: 10.1016/j.bbrc.2019.12.051. PMID: 31862139
15. Xu X, **Kobayashi S**, Timm D, Huang Y, Zhao F, Shou W, Liang Q. (2019) Enhanced mTOR complex 1 signaling attenuates diabetic cardiac injury in OVE26 mice. *FASEB J*. PMID: 31469601. Published Online:30 Aug 2019. Doi: 10.1096/fj.201901206R. PMID: 31469601
16. Catanzaro MP, Weiner A, Kaminaris A, Li C, Cai F, Zhao F, **Kobayashi S**, Kobayashi T, Huang Y, Liang Q. Doxorubicin-induced cardiomyocyte death is mediated by unchecked mitochondrial fission and mitophagy. *FASEB J*. Published Online:10 Jul 2019. doi: 10.1096/fj.201802663R. PMID: 31291545
17. Ying M, Yasuda H, **Kobayashi S**, Sakurai N, Kidou S. (2019) Barley cold-induced CISP proteins contribute to the accumulation of heavy metals in roots. *Environmental and Experimental Botany*. Volume 165, September 2019, Pages 53-58
18. Yokoyama Y, **Kobayashi S**, Kidou S. PHD type zinc finger protein PFP represses flowering by modulating FLC expression in *Arabidopsis thaliana*. *Plant Growth Regulation*. Epub 2019 Mar 18. doi: 10.1007/s10725-019-00487-1.
19. Klionsky DJ, et al. Guidelines for the use and interpretation of assays for monitoring autophagy (3rd edition). *Autophagy*. 2016;12(1):1-222. doi: 10.1080/15548627.2015.1100356. PMID 26799652.
20. Liang Q, **Kobayashi S**. Mitochondrial quality control in the diabetic heart. *J Mol Cell Cardiol*. 2015 Dec 29. pii: S0022-2828(15)30163-2. doi: 10.1016/j.yjmcc.2015.12.025. PMID 26739215 (Review article)
21. **Kobayashi S**. Choose Delicately and Reuse Adequately: The Newly Revealed Process of Autophagy. *Biol Pharm Bull*. 2015;38(8):1098-103. doi: 10.1248/bpb.b15-00096. PMID 2623572 (Review article)

22. **Kobayashi S**, Liang Q. Autophagy and mitophagy in diabetic cardiomyopathy Biochim Biophys Acta. Epub 2014 May 29. doi: 10.1016/j.bbadis.2014.05.020. PMID 24882754 (Review article)
23. **Kobayashi S**, Xu X, Chen K, Timm D, Volden P, Gulick J, Yue Z, Robbins J, Epstein PN, and Liang Q. Diminished Autophagy Limits Cardiac Injury in Mouse Models of Type 1 Diabetes. J Biol Chem. 2013, Epub2013 May 8. doi: 10.1074/jbc.M113.474650.
24. Chen K, Xu X, **Kobayashi S**, Viollet B, and Liang Q. AMP Activated Protein Kinase Is Indispensable for Myocardial Adaptation to Caloric Restriction in Mice. PLoS ONE 2013 8(3): e59682. Epub 2013 Mar 19. doi: 10.1371/journal.pone.0059682
25. Ounzain S, **Kobayashi S**, Peterson RE, He A, Motterle A, Samani NJ, Menick DR, Pu WT, Liang Q, Chong NW. Cardiac Expression of ms1/STARS, a Novel Gene Involved in Cardiac Development and Disease is regulated by GATA4. 2012 32(10):1830-43. Epub 2012 Mar 19. doi: 10.1128/MCB.06374-11
26. **Kobayashi S**, Xu X, Chen K, and Liang Q. Suppression of Autophagy Is Protective in High Glucose-Induced Cardiomyocyte Injury. Autophagy 2012 Apr; 8(4) 1-16. doi: 10.4161/auto.18980
27. Xu X, Chen K, **Kobayashi S**, Timm D, and Liang Q. RESVERATROL ATTENUATES DOXORUBICIN-INDUCED CARDIOMYOCYTE DEATH VIA INHIBITION OF S6K1-MEDIATED AUTOPHAGY. J Pharmacol Exp Ther 2012, Epub 2011 Dec 30. doi: 10.1124/jpet.111.189589
28. Chen K, Xu X, **Kobayashi S**, Timm D, Jepperson T, and Liang Q. Caloric Restriction Mimetic 2-Deoxy-Glucose Antagonizes Doxorubicin-Induced Cardiomyocytes Death through Multiple Mechanisms. J Biol Chem. 2011 Jun; 286(25):21993-2006, Epub 2011 Apr 26. doi: 10.1074/jbc.M111.225805
29. **Kobayashi S**, Volden P, Timm D, Mao K, Xu X, and Liang Q. Transcription Factor GATA4 Inhibits Doxorubicin-Induced Autophagy and Cardiomyocyte Death. J Biol Chem. 2010 Jan; 285(1):793-804. Epub 2009 Nov 9. doi: 10.1074/jbc.M109.070037
30. Huang Y, Wright CD, **Kobayashi S**, Healy CL, Eglethun M, Cypher AL, Liang Q, and O'Connell TD. GATA4 is a Survival Factor in Adult Cardiac Myocytes, but Is Not required for alpha1A-Adrenergic Receptor Survival Signaling. Am J Physiol 2008 295: H699-H707. Epub 2008 Jun 13. doi: 10.1152/ajpheart.01204.2007
31. **Kobayashi S**, Mao K, Huang Y, Jaffer ZM, Chernoff J, Volden P, and Liang Q. Regulation of Akt/PKB activity by P21 activated kinase in cardiac myocytes. J Mol Cell Cardiol. 2008 Feb; 44(2):429-34. Epub 2007 Dec 3.

32. Chen Y, **Kobayashi S**, Redetzke R, Said S, Liang Q, and Gerdes AM. Short-term Triiodo-L-Thyronine treatment inhibits cardiac myocyte apoptosis in border area after myocardial infarction in rats. *J Mol Cell Cardiol*. 2008 Jan; 44(1):180-7. Epub 2007 Oct 26.
33. **Kobayashi S**, Mao K, Zheng H, Wang X, Patterson C, O'Connell TD and Liang Q. Diminished GATA4 Protein Levels Contribute to Hyperglycemia-induced Cardiomyocyte Injury. *J Biol Chem*. 2007 Jul 27; 282(30):21945-21952. Epub 2007 May 24.
34. **Kobayashi S**, Lackey T, Huang Y, Bisping E, Pu WT, Boxer LM, and Liang Q. Transcription Factor Gata4 Regulates Cardiac Bcl2 Gene Expression *In Vitro* And *In Vivo*. *FASEB J*. 2006 Apr; 20(6):800-2. Epub 2006 Feb 9.
35. Kuzman JA, Gerdes AM, **Kobayashi S**, and Liang Q. T3 Activates AKT and Prevents Serum Starvation Induced Cell Death in Neonatal Cardiomyocytes. *J Mol Cell Cardiol*. 2005 Nov; 39(5):841-4.
36. Rao D, Momcilovic I, **Kobayashi S**, Callegari E, and Ristic Z. Chaperone Activity of Recombinant Maize Chloroplast Protein Synthesis Elongation Factor, EF-Tu. *Eur J Biochem* 2004 271(18): 3684-92.
37. Ristic Z, Wilson K, Nelsen C, Momcilovic I, **Kobayashi S**, Meeley R, Muszynski M, Habben J. A maize mutant with decreased capacity to accumulate chloroplast protein synthesis elongation factor (EF-Tu) displays reduced tolerance to heat stress. *Plant Science* 2004 167:1367-1374.
38. Kamiie K, Nomura Y, **Kobayashi S**, Taira H, Kobayashi K, Yamashita T, Kidou S, and Ejiri S. Cloning and expression of Bombyx mori silk gland elongation factor 1 γ in *Escherichia coli*. *Biosci Biotechnol Biochem*. 2002 Mar;66(3):558-65.
39. **Kobayashi S**, Kidou S, Ejiri S. Detection and characterization of glutathione S-transferase activity in rice EF-1 β and EF-1 γ expressed in *Escherichia coli*. *Biochem Biophys Res Commun*. 2001 Nov 2;288(3):509-14.
40. Kidou S, Tsukamoto S, **Kobayashi S**, and Ejiri S. Isolation and characterization of a rice cDNA encoding the gamma-subunit of translation elongation factor 1B (eEF1B γ). *FEBS Lett*. 1998 Sep 4;434(3):382-6.

Complete List of Published Work in MyBibliography:

<http://www.ncbi.nlm.nih.gov/sites/myncbi/satoru.kobayashi.1/bibliography/40433448/public/>

b) Non-peer reviewed article

c) Manuscripts that are currently under consideration/review by a scientific/medical journal but are not yet accepted

Lionel Chong, Hema Rampersaud, Anamika Rao, Isha Gujare, Amrita Makhijani, Dokyeong Kim, Elizabeth Levy, Brian Beatty, Satoru Kobayashi.

Gut–Heart–Kidney Communication in Cardiovascular Disease: From a Single-Metabolite Hypothesis to an Orchestrating Metabolite-Network Paradigm. (Review Article) Am J Physiol Heart Circ Physiol. Under revision.

Teresa Zheng, Anushka Bhatt, Helen Deng, Joshua Pribadi, Kieran Mannion, Naman Kapoor, Heba Ahmed, Sabrina Martinez, Satoru Kobayashi

The Role of Lysosomes in Doxorubicin-Induced Cardiotoxicity. (Review Article) In preparation.

Naman Kapoor, Heba Ahmed, Brett Silverstein, Louis Cui, Daniel Nasef, Demarcus Nasef, Tamayo Kobayashi, Yuan Huang, Lars Udo-Bellner, Amanda Charest, Randy Stout, Qiangrong Liang, Satoru Kobayashi. Imidazoline Receptor Agonists Alleviate Doxorubicin-Induced Lysosomal Injury and Cardiomyocyte Death. (Research Article) In preparation.

Books and book chapters (in descending chronological order)

None

Published abstracts and meeting presentations (list in descending chronological order)

1. Jack Chin, Sabrina Martinez, Tamayo Kobayashi, Qiangrong Liang, **Satoru Kobayashi**. Doxorubicin induced differential response of lysosome-associated proteins in human cardiomyocytes. Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2026, Jul 13–Jul 16, 2026, Boston, MA.
2. Jack Chin, Sabrina Martinez, Tamayo Kobayashi, Qiangrong Liang, **Satoru Kobayashi**. Doxorubicin induced differential response of lysosome-associated proteins in human cardiomyocytes. Asian Cardiovascular Symposium (Pre-BCVS) 2026, Jul 12, 2026, Boston, MA.
3. Ronald James Lospenuso III, Vignesh Gunasekaran, Shilpreet Brar, Aila Sajjad, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. AMPK γ 1 and γ 2 Isoforms Protect Against Doxorubicin-Induced Cardiotoxicity via Autophagy Regulation. Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2026, Jul 13–Jul 16, 2026, Boston, MA.
4. **Satoru Kobayashi**, Lars Udo-Bellner, Amanda Charest, Kieran Mannion, Jack Chin, Chandrema Hossain, Vignesh Gunasekaran, Shilpreet Brar, Elizabeth Levy, Yuan Huang, Tamayo Kobayashi, Qiangrong Liang, Randy Stout. Live-cell imaging reveals altered mitochondria-lysosome contact dynamics in doxorubicin-induced cardiotoxicity. American Physiology Summit 2026, April 23-26, 2026, Minneapolis, MN (Oral and poster)

5. Heba Ahmed, Brett Silverstein, Naman Kapoor, Rini Lee, Louis Cui, Tamayo Kobayashi, Yuan Huang, Amanda Charest, Randy Stout, Qiangrong Liang, **Satoru Kobayashi**. Ubiquitin E3 ligase SMURF1 Alleviates Doxorubicin-Induced Lysosomal Injury and Cardiomyocyte Death. Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2025, Jul 23–Jul 26, 2025, Baltimore, MD.
6. Naman Kapoor, Heba Ahmed, Brett Silverstein, Louis Cui, Daniel Nasef, Demarcus Nasef, Tamayo Kobayashi, Yuan Huang, Amanda Charest, Randy Stout, Qiangrong Liang, **Satoru Kobayashi**. Imidazoline Receptor Agonists Alleviate Doxorubicin-Induced Lysosomal Injury and Cardiomyocyte Death. American Physiology Summit 2025, April 24-27, 2025, Baltimore, MD.
7. Raymond Wong, Pravin Vathappallil, Alessia Vata, Khevana Patel, Tamayo Kobayashi, Qiangrong Liang, **Satoru Kobayashi**. The imidazoline I2 receptor agonist 2-BFI attenuates Glucolipotoxicity in cardiomyocytes. American Physiology Summit 2025, April 24-27, 2025, Baltimore, MD.
8. Christopher Lau, Anushka Bhatt, Pritpaul Singh, Chris Mathew, **Satoru Kobayashi**, Qiangrong Liang. Time-Dependent Autophagy Flux and Cell Death Responses in Cardiomyocytes Treated with Doxorubicin. American Physiology Summit 2025, April 24-27, 2025, Baltimore, MD.
9. Pritpaul Singh, Chris Mathew, Emily Joseph, Christopher Lau, **Satoru Kobayashi**, Qiangrong Liang. P21-activated kinase 1 (PAK1) Regulates Autophagy in Human Cardiomyocytes. American Physiology Summit 2025, April 24-27, 2025, Baltimore, MD.
10. Anushka Bhatt, Christopher Lau, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. Autophagy Flux and Cell Death Responses in Doxorubicin Treated Cardiomyocytes. Medical Society of the State of New York (MSSNY) Poster Symposium, April 4, 2025, Tarrytown, New York
11. **Satoru Kobayashi**, Demarcus Nasef, Daniel Nasef, Heba Ahmed, Naman Kapoor, Young Hee Hahn, Tamayo Kobayashi, Yuan Huang, Randy Stout, Qiangrong Liang. Redefining the Role of lysosomal membrane permeabilization (LMP) in Doxorubicin-induced cardiotoxicity. The 3rd NYC Inter-Institutional Cardiovascular Retreat, Sep 26, 2024, Old Westbury, NY. (selected oral presentation)
12. Church I, Van J, Silverstein B, Chen H, Kobayashi T, **Kobayashi S**, Liang Q. AMPK β 1 knockdown increases autophagy flux and Doxorubicin-induced H9c2 cardiomyocyte death. American Osteopathic Association (AOA) OMED, September 19-22, 2024, San Antonio, Texas
13. John Yarbrow, Lionel Chong, Caleb Sooknanan, Chris Matthew, Pritpaul Singh, Michelle Kim, Kinza Ahmed, Rohan Francis, Tamayo Kobayashi, Yuan Huang, **Satoru Kobayashi**, Qiangrong Liang. P21-activated kinase 1 mediated autophagy regulation: A safeguard from

cardiac impairment during fasting? Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2024, Jul 22–Jul 25, 2024, Chicago, IL.

14. Demarcus Nasef, Daniel Nasef, Younghee Hahn, Tamayo Kobayashi, Yuan Huang, Randy Stout, Qiangrong Liang, **Satoru Kobayashi**. Doxorubicin-Induced Cardiomyocyte Death is mediated by Lysosomal Membrane Permeabilization. Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2024, Jul 22–Jul 25, 2024, Chicago, IL.
15. Lionel Chong, BS, Caleb Sooknanan, Shawn Geffken, Tint Tha Ra Wun, Anoushka Guha, Pooja Jaiswal, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. PAK1 regulates cardiomyocyte autophagy through p62. Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2024, Jul 22–Jul 25, 2024, Chicago, IL.
16. Lionel Chong, BS, Caleb Sooknanan, Shawn Geffken, Tint Tha Ra Wun, Anoushka Guha, Pooja Jaiswal, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. PAK1 regulates cardiomyocyte autophagy through p62. The 6th Asian Cardiovascular Symposium (ACS), Jul 21, 2024, Chicago, IL.
17. Nidhi Banker, Pravin Vathappallil, Anushka Bhatt, Tamayo Kobayashi, Qiangrong Liang, **Satoru Kobayashi**. Cardiomyocytes Enhance Lysosomal Function to Mitigate Hyperlipidemia-Induced Cell Death. The Symposium of University Research and Creative Expression (SOURCE), Apr 14, 2024, Manhattan, NY
18. Younghee Hahn, Hongling Chen, Nidhi Banker, Srinidhi Gadula, Pravin Vathappallil, Brett Silverstein, Jennifer Van, Tamayo Kobayashi, Ziyang Zhang, Yuan Huang, Qiangrong Liang, **Satoru Kobayashi**. Glucolipotoxicity Curtails Mitophagy In Cardiomyocytes. Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2023, Jul 31–Aug 3, 2023, Boston, MA.
19. Shawn Geffken, Tint TR Wun, Anoushka Guha, Pooja Jaiswal, Kinza Ahmed, Caleb Sooknanan, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. P62 Overexpression Rescues Defective Autophagy Flux In Cardiomyocytes Deficient In P21 Activated Kinase 1. Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2023, Jul 31–Aug 3, 2023, Boston, MA.
20. Younghee Hahn, Hongling Chen, Nidhi Banker, Srinidhi Gadula, Pravin Vathappallil, Brett Silverstein, Jennifer Van, Tamayo Kobayashi, Ziyang Zhang, Yuan Huang, Qiangrong Liang, **Satoru Kobayashi**. Glucolipotoxicity Curtails Mitophagy In Cardiomyocytes. Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2023, Jul 31–Aug 3, 2023, Boston, MA.
21. Shawn Geffken, Tint TR Wun, Anoushka Guha, Pooja Jaiswal, Kinza Ahmed, Caleb Sooknanan, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. P62 Overexpression Rescues Defective Autophagy Flux In Cardiomyocytes Deficient In P21 Activated Kinase 1. Basic Cardiovascular Sciences (BCVS) Scientific Sessions 2023, Jul 31–Aug 3, 2023, Boston, MA.

22. Jennifer Van, Lokesh Katiki, Rahul Patel, Max Cohen, Puja Mehta, Yawen Zhang, Katherine Livatova, Salman Alam, Jaclyn Del Pozzo, Tamayo Kobayashi, Yuan Huang, **Satoru Kobayashi**, Qiangrong Liang. Metformin Inhibits Autophagy/Mitophagy and Antagonizes Doxorubicin-Induced Cardiomyocyte Death. The American Heart Association Scientific Sessions 2022, Chicago, IL Nov. 5-7 2022
23. Hongling Chen, Younghee Hahn, Brett Silverstein, Tamayo Kobayashi, Yuan Huang, **Satoru Kobayashi**, Qiangrong Liang. High glucose and high fat differentially regulate autophagic flux in H9C2 cardiac myoblasts, Cell Bio 2022-An ASCB|EMBO meeting 2022, Washington, DC, Dec. 3-7 2022
24. Shawn Geffken, Anoushka Guha, Pooja Jaiswai, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. An in-Depth Analysis of the PAK1 Autophagy Signaling Pathway in H9C2 Cardiomyoblasts. Experimental Biology 2022 Philadelphia, PA April 2-5, 2022
25. Mandeep Singh, Naunihal Singh, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. Increased Connexin 43 expression in AMPK deficient cardiomyocytes in response to Doxorubicin treatment. Experimental Biology 2022 Philadelphia, PA April 2-5, 2022
26. Christopher Chen, Tony Nguyen, **Satoru Kobayashi**, Qiangrong Liang. The dichotomous function of FUNDC1-dependent mitophagy in doxorubicin cardiotoxicity. Experimental Biology 2022 Philadelphia, PA April 2-5, 2022
27. Mandeep Singh, Naunihal Singh, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. Connexin 43 Up-regulation by AMPK Deficiency in Cardiomyocytes Treated With the Anticancer Drug Doxorubicin. American Osteopathic Association (AOA) Osteopathic Medical Conference and Exposition (OMED) Conference, Virtual conference, October 22-24, 2021.
28. Vignesh Gunasekaran, Naunihal Singh, Tamayo Kobayashi, Qiangrong Liang, **Satoru Kobayashi**. The differential roles of AMPK γ subunit isoforms in doxorubicin-induced H9c2 cell death. American Osteopathic Association (AOA) Osteopathic Medical Conference and Exposition (OMED) Conference. October 22-24, 2021
29. Shawn Geffken, Tint Tha Ra Wun, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. P21-Activated Kinase 1 is Essential for the Gene Expression of the Autophagy Adaptor P62/SQSTM1 in Cardiomyocytes. American Osteopathic Association (AOA) Osteopathic Medical Conference and Exposition (OMED) Conference. October 22-24, 2021
30. Akito Nicol, Mandeep Singh, Jaclyn Del Pozzo, Tint Tha Ra Wun, Puja Mehta, Tamayo Kobayashi, **Satoru Kobayashi**, Qiangrong Liang. Knockdowns of AMPK α Subunit Isoforms Result in Differential H9c2 Cell Responses to Doxorubicin Treatment. Experimental Biology 2021. Apr 27-30, 2021. (Virtual format)

31. Lauren Soong, Katrina Bantis, **Satoru Kobayashi**, Youhua Zhang, Qiangrong Liang. FUNDC1 is Essential for Maintaining Cardiac Function During Prolonged Fasting. Experimental Biology 2021. Apr 27-30, 2021. (Virtual format)
32. Katrina Bantis, **Satoru Kobayashi**, Youhua Zhang, Qiangrong Liang. FUNDC1 is Required for the Maintenance of Normal Cardiac Function During Fasting. American Osteopathic Association (AOA) Osteopathic Medical Conference and Exposition (OMED) Conference, Baltimore, MD. October 25-28, 2019.
33. Katrina Bantis, **Satoru Kobayashi**, Youhua Zhang, Qiangrong Liang. The Role of FUNDC1 in the Heart During Fasting. 2019 NHLBI Mitochondrial Biology Symposium, Bethesda, ML. September 26-27, 2019.
34. Joy Patel, **Satoru Kobayashi**, Tamayo Kobayashi, Qiangrong Liang. Impaired Mitophagy Leads to Cardiac Dysfunction in Diabetic and High Fat Fed Mice. 2019 NHLBI Mitochondrial Biology Symposium, Bethesda, ML. September 26-27, 2019.
35. Jaclyn Del Pozzo, Puja Mehta, Fei Cai, Cairong Li, **Satoru Kobayashi**, and Qiangrong Liang. Unexpected Role of Adenosine-Monophosphate Activated Protein Kinase in Doxorubicin Cardiotoxicity and Metformin-Mediated Cardioprotection. The 23rd Annual Scientific Meeting of Heart failure Society of America (HFSA), Philadelphia, PA. September 13-15, 2019. (One of six finalists for oral presentation in the Jay N. Cohn New Investigator Award: Basic Science Session)
36. Bharvi Marsha Patel, Anam Chaudhry, **Satoru Kobayashi**, and Qiangrong Liang. Transcription Factor EB and It's Potential Protective Role in Hyperglycemic Cardiotoxicity. The 23rd Annual Scientific Meeting of Heart failure Society of America (HFSA), Philadelphia, PA. September 13-15, 2019.
37. Qiangrong Liang, Fengyi Zhao, Tamayo Kobayashi, **Satoru Kobayashi**, Yuan Huang. An impaired mitophagy-lysosome system contributes to hyperglycemia cardiotoxicity. The American Diabetes Association's (ADA's) 79th Scientific Sessions 2019 Jun 7 - 11, 2019, San Francisco, CA.
38. Omar Nagaria, Suprit Singh, Rizowana Kabir, **Satoru Kobayashi**, Tamayo Kobayashi, Qiangrong Liang. Kinetin is Sufficient to Accelerate Mitophagy Flux in H9c2 Cardiac Myoblast Cells Experimental Biology 2019 Apr 9 - 2019, Los Angeles, CA.
39. Yawen Zhang, Salman Alam, Tamayo Kobayashi, Fengyi Zhao, Yuan Huang, **Satoru Kobayashi**, Qiangrong Liang. Differential Effects of Beclin1 on Autophagy and Mitophagy in Cardiomyocytes. AOA-OMED Conference 2018 Oct 5-Oct 9 2018, San Diego, CA.
40. Katrina Bantis, Jeremy A Garcia, **Kobayashi S**, and Liang Q. Prolonged fasting attenuates mitophagy and impairs cardiac function in mice. AOA-OMED Conference 2017 Oct 7-Oct 10 2017, Philadelphia, PA. (Student Poster Competition 1st Prize Winner)

41. Jeremy A Garcia, Katrina Bantis, Usman Aslam, **Kobayashi S**, and Liang Q. The Effect of Fasting on Mitochondrial Size and Mitophagy in Cardiomyocytes. AOA-OMED Conference 2017 Oct 7-Oct 10 2017, Philadelphia, PA.
42. Michael Chang, Yawen Zhang, **Kobayashi S**, and Liang Q. The Effect of Fasting on Mitochondrial Size and Mitophagy in Cardiomyocytes. Experimental Biology 2018 Apr 15-18 2018, San Diego, CA. (Nominated as a poster award finalist for an AAA Graduate Student Poster Award at AAA's Annual Meeting at EB 2018)
43. Kevin Clements, Kathy Tin, Tamayo Kobayashi, Yuan Huang, **Kobayashi S**, and Liang Q. P21-Activated Kinase 1 is Necessary for Autophagy and Mitophagy in Cardiomyocytes. Experimental Biology 2018 Apr 15-18 2018, San Diego, CA.
44. Katrina Bantis, Yawen Zhang, Tamayo Kobayashi, Yuan Huang, **Kobayashi S**, and Liang Q. Mitophagy Must Be Maintained at a Certain Level in Order to Provide Cardioprotection in Mice During Fasting. Experimental Biology 2018 Apr 15-18 2018, San Diego, CA.
45. Kaminaris A, **Kobayashi S**, McStay G, and Liang Q. AMPK Negatively Regulates Mitophagy in the Heart. The American Society of Biochemistry and Molecular Biology, Experimental Biology 2017, April 22-26, 2017, Chicago, IL. (ASBMB Graduate/Postdoctoral Travel Award—2017 Recipient (\$1,000), The American Society of Biochemistry and Molecular Biology, Experimental Biology 2017)
46. Qiangrong Liang, **Satoru Kobayashi**, Fengyi Zhao, Yuan Huang, Tamayo Kobayashi. Parkin-Mediated Mitophagy Is Essential for Maintaining Cardiac Function in Diabetic Mice. The American Diabetes Association 77th Scientific Sessions, June 9-13, 2017, San Diego, CA.
47. Qiangrong Liang, **Satoru Kobayashi**, Fengyi Zhao, Usman Aslam, Katrina Bantis, Eric Lam, Yuan Huang, Tamayo Kobayashi, Amanda Kaminaris. Differential Roles of Mitophagy in the Heart under Fasting and Diabetic Conditions. INTERNATIONAL ACADEMY OF CARDIOLOGY ANNUAL SCIENTIFIC SESSIONS 2017, 22nd WORLD CONGRESS ON HEART DISEASE, VANCOUVER, BC, CANADA, JULY 14-16, 2017.
48. **Satoru Kobayashi**, Yuan Huang, Tamayo Kobayashi, Qiangrong Liang. Lysosomal Membrane Permeabilization Contributes to the Hyperglycemia-Induced Cardiomyocyte Dysfunction. The American Heart Association, Basic Cardiovascular Sciences 2017 Scientific Sessions, July 10-13 2017, Portland, OR.
49. Fang Li, Yuan Huang, **Satoru Kobayashi**, Qiangrong Liang, Ioana Voiculescu. Cell-substrate impedance sensing (ECIS) for evaluation of drug-induced cardiovascular toxicity using iPSC-derived cardiomyocyte. Stem Cells for Drug Discovery & Toxicity Screening 2017, July 10-11, Boston, MA.
50. **Satoru Kobayashi**, Yuan Huang, Tamayo Kobayashi, Fengyi Zhao, Qiangrong Liang. Lysosomal Membrane Permeabilization and The Ensuing Ectopic Expression of Cathepsin D

Mediates Hyperglycemia-Induced Cardiomyocyte Death. Annual Meeting of the International Academy of Cardiovascular Sciences (IACS): North American Section. 5th Annual Forum to Promote Young Investigators and Centers of Excellence in Cardiovascular Sciences. August 31-September 2 2017, Orlando, FL.

51. Weiner A, Kaminaris A, **Kobayashi S**, and Liang Q. Mitophagy Contributes to Doxorubicin-Induced Cardiomyocyte Death. American College of Physicians, Internal Medicine Meeting 2016, May 5-7 2016, Washington D.C. (Young Achiever/Abstract Finalist in ACP's National Abstract Competition)
52. Weiner A, Kaminaris A, **Kobayashi S**, Gerdes M, and Liang Q. The Role of Mitophagy in Doxorubicin-Induced Cardiomyocyte Death. Experimental Biology Annual Meeting, April 2-7, San Diego, CA
53. Aslam U, **Kobayashi S**, and Liang Q. Time-Dependent Differential Effects of Fasting on Cardiac Autophagy and Mitophagy. The American Physiological Society, Experimental Biology 2016, April 2-6, 2016, San Diego, CA. (Excellent in Professional Student Research Travel Award—2016 Recipient (\$1800), The American Physiological Society, Experimental Biology 2016)
54. Chirag P, **Kobayashi S**, and Liang Q. The American Physiological Society, Experimental Biology 2016, April 2-6, 2016, San Diego, CA. (Excellent in Professional Student Research Travel Award—2016 Recipient (\$1800), The American Physiological Society, Experimental Biology 2016)
55. Erstenyuk, V., **Kobayashi, S.**, and Liang, Q. GATA4 Reduces Doxorubicin-induced Cardiomyocyte Death by Inhibiting Mitophagy. New York State Osteopathic Medical Society Scientific Society Poster Competition, April 01 2016, Hauppauge, NY
56. Nuamah J, **Kobayashi S** and Liang Q, Effects of 2DG on Mitochondrial mitophagy in Doxorubicin treated cardiomyocyte. New York State Osteopathic Medical Society Scientific Society Poster Competition, April 01 2016, Hauppauge, NY
57. Benjamin P Kramer, **Kobayashi S**, and Liang Q. High Glucose Limits Autophagy via Inhibition of Nuclear Translocation of Transcription Factor EB. AOA-OMED Conference 2015 Oct 17-Oct 21 2015, Orland, FL. (Student Poster Competition 2nd Prize Winner).
58. **Kobayashi S**, Gilliland M.G.F, Hoffman W.H, O'Connor A, Jung MK, and Liang Q. Diabetic Ketoacidosis Compromises Cardiac Mitochondrial Quality Control in Humans. The American Heart Association, Basic Cardiovascular Sciences 2015 Scientific Sessions, July 13-16 2015, New Orleans, LA.
59. **Kobayashi S**, Gilliland M.G.F, Hoffman W.H, O'Connor A, Jung MK, and Liang Q. Diabetic Ketoacidosis Compromises Cardiac Mitochondrial Quality Control in Humans. The Second Annual ACRE/APS Symposium: Hot Topics in Current Basic and Translational

Research of CV disease, July 12, 2015, New Orleans, LA. (prized for Outstanding Research Award)

60. Catanzaro M.P, Weiner A, Kaminaris A, **Kobayashi S**, and Liang Q. Doxorubicin-induced H9c2 Cell Death is Mediated by Excessive Mitochondrial Fission and Mitophagy. The American Heart Association, Basic Cardiovascular Sciences 2015 Scientific Sessions, July 13-16 2015, New Orleans, LA.
61. Catanzaro M.P, Weiner A, Kaminaris A, **Kobayashi S**, and Liang Q. Doxorubicin-induced H9c2 Cell Death is Mediated by Excessive Mitochondrial Fission and Mitophagy. The Second Annual ACRE/APS Symposium: Hot Topics in Current Basic and Translational Research of CV disease, July 12, 2015, New Orleans, LA. (Outstanding Research Award)
62. **Kobayashi S**, Gilliland M.G.F, Hoffman W.H, O'Connor A, and Liang Q. Dysregulation of Cardiac Mitochondrial Quality Control in Young Adults and Adolescents with Fatal Diabetic Ketoacidosis. . The American Diabetes Association, the 75th Scientific Sessions, June 12-17 2015, Boston, MA.
63. Michael P. Catanzaro, **Kobayashi S**, and Liang Q. Mitochondrial Fragmentation and Mitophagy Contribute to Doxorubicin-induced Cardiomyocyte Death. Experimental Biology meeting EB2015 Mar 28-Apr 1 2015, Boston, MA. (Travel Award)
64. **Kobayashi S**, Huang Y, and Liang Q. Mitophagy in Type 1 Diabetic Heart. The Keystone Symposia on Molecular and Cellular Biology, (J15) Mitochondria, Metabolism and Heart Failure (J5) Jan27-Feb 1 2015, Santa Fe, NM.
65. **Kobayashi S**, Huang Y, and Liang Q. High Glucose-Induced Mitophagy Limits Cardiomyocyte Injury. The American Heart Association, Basic Cardiovascular Sciences 2014 Scientific Sessions, July 14-17 2014, Las Vegas, NV.
66. **Kobayashi S**, Huang Y, and Liang Q. High Glucose-Induced Mitophagy Limits Cardiomyocyte Injury. The First Annual ACRE/APS Symposium: Novel Therapeutic Approaches, July 13, 2014, Las Vegas, NV.
67. Xu X, **Kobayashi S**, Chen K, Timm D Shou W, and Liang Q. Enhanced mTOR signaling attenuates cardiac injury in OVE26 diabetic mice. The American Diabetes Association, the 74th Scientific Sessions, June 13-17 2014, San Francisco, CA.
68. **Kobayashi S**, Xu X, Chen K, Timm D and Liang Q. Mitochondrial Autophagy Prevents Hyperglycemia-Induced Cardiomyocyte Injury. The American Diabetes Association, the 73rd Scientific Sessions, June 21-25 2013, Chicago, IL.
69. Chen K, **Kobayashi S**, Xu X, and Liang Q. AMP Activated Protein Kinase Is Indispensable for Maintaining Normal Cardiac Function during Caloric Restriction. The American Heart Association, Scientific Sessions, November 3-7 2012, Los Angeles, CA.

70. Chen K, Xu X, **Kobayashi S**, Timm D and Liang Q. Autophagy is Essential for Enhanced Heart Function by Swimming Exercise. The American Heart Association, Scientific Sessions, November 3-7 2012, Los Angeles, CA.
71. **Kobayashi S**, Xu X, Chen K, Timm D and Liang Q. Cathepsin D Accumulation is Responsible for Hyperglycemia-Induced Angiotensin II Synthesis in Cardiomyocytes. The American Diabetes Association, the 72nd Scientific Sessions, June 8-12 2012, Philadelphia, PA.
72. Xu X, **Kobayashi S**, Chen K, Timm D, Volden P and Liang Q. The American Diabetes Association, the 72nd Scientific Sessions, June 8-12 2012, Philadelphia, PA.
73. Chen K, Timm D, Jepperson T, Xu X, **Kobayashi S** and Liang Q. Necessity of mTOR Inhibition and Autophagy Induction for Maintaining Cardiac Homeostasis during Caloric Restriction. The American Heart Association, Basic Cardiovascular Sciences 2011 Scientific Sessions, July 2011, New Orleans, LA.
74. **Kobayashi S**, Xu X, Saso-Hagiwara M, Chen K, Timm D, Jepperson T and Liang Q. Cathepsin Accumulations Contribute To the Hyperglycemia-Induced Cardiomyocyte Injury. The American Diabetes Association, the 71st Scientific Sessions, June 2011, San Diego, CA.
75. Xu X, **Kobayashi S**, Chen K, Timm D, Volden P and Liang Q. Diminished Autophagy is an Adaptive Response that Protects against Diabetic Cardiac Injury The American Diabetes Association, the 71st Scientific Sessions, June 2011, San Diego, CA.
76. **Kobayashi S**, Volden P, Timm D, Jepperson T and Liang Q. GATA4 Antagonizes Doxorubicin-induced Myocardial Injury by Suppressing Autophagy. The American Heart Association, Basic Cardiovascular Sciences 2010 Scientific Sessions, July 2010, Rancho Mirage, CA. (This abstract was judged as among the top 10% of accepted abstracts and invited to present at the AHA Scientific Sessions in Chicago November 2010 in the “Best of the AHA Specialty Conferences Poster Session”)
77. Chen K, Xu X, **Kobayashi S**, Timm D, Jepperson T and Liang Q. 2-Deoxy-Glucose Protects from Doxorubicin-induced Cardiomyocyte Death through Activation of AMP-activated Protein Kinase and Inhibition of Autophagy. The American Heart Association, Basic Cardiovascular Sciences 2010 Scientific Sessions, July 2010, Rancho Mirage, CA. (This abstract was judged as among the top 10% of accepted abstracts and invited to present at the AHA Scientific Sessions in Chicago November 2010 in the “Best of the AHA Specialty Conferences Poster Session”)
78. **Kobayashi S**, Chen K, Xu X, Timm D, Liang Q. Inhibited Autophagy Is Protective in Hyperglycemia-Induced Cardiomyocyte Injury. 70th Scientific Sessions, American Diabetes Association, June 2010, Orlando, FL.
79. Xu X, Chen K, **Kobayashi S**, Timm D, Liang Q. GATA4 is essential for cardiac homeostasis in streptozotocin-induced diabetes. 70th Scientific Sessions, American Diabetes Association, June 2010, Orlando, FL. (This poster was selected for the audio poster tour)

80. **Kobayashi S**, Volden P, Timm D, Mao K and Liang Q. GATA4 Protects from Doxorubicin Cardiotoxicity through Inhibition of Autophagy. 6th Annual Symposium of the American Heart Association Council on Basic Cardiovascular Sciences, July 2009, Las Vegas, NV.
81. **Kobayashi S**, Mao K, Volden P, Tricia K, Troy L and Liang Q. GATA4 Inhibits Doxorubicin-induced Autophagy and Protects from Cardiomyocyte Death. 5th Annual Symposium of the American Heart Association Council on Basic Cardiovascular Sciences, July 2008, Keystone, CO.
82. **Kobayashi S**, Mao K, Volden P, Tricia K, Troy L and Liang Q. Inhibition of Autophagy by GATA4 Protects from Doxorubicin-induced Cardiomyocyte death. International Society for Heart Research North America Section Annual meeting, June 19, 2008, Cincinnati, IL
83. **Kobayashi S**, Mao K, Huang Y, Wang X, O'Connell TD, Liang Q. Novel Mechanisms of Anthracycline-induced Cardiotoxicity. American Heart Association Research Symposium, Nov 3, 2007, Orlando, FL
84. **Kobayashi S**, Mao K, Huang Y, Jaffer ZM, Chernoff J, and Liang Q. Regulation of Akt/PKB activity by P21 activated kinase in cardiac myocytes. 4th Annual Symposium of the American Heart Association Council on Basic Cardiovascular Sciences, July-August 2007, Keystone, CO.
85. Mao K, **Kobayashi S**, Zheng H, Wang X, Patterson C, O'Connell TD and Liang Q. Diminished GATA4 Protein Levels Contribute to Hyperglycemia-induced Cardiomyocyte Injury. 4th Annual Symposium of the American Heart Association Council on Basic Cardiovascular Sciences, July-August 2007, Keystone, CO.
86. **Kobayashi S**, Huang Y, Lackey T, Liu J, Wang X, O'Connell T, Liang Q. Abnormal Activation of the Ubiquitin Proteasome System is a Novel Mechanism that Mediates Doxorubicin-induced Cardiotoxicity. 3rd Annual Symposium of the American Heart Association Council on Basic Cardiovascular Sciences, August 2006, Keystone, CO.
87. Liang Q, **Kobayashi S**, Huang Y, Lackey T, Wang X, O'Connell TD. Activation of Ubiquitin Proteasome System Contributes to Doxorubicin-induced Cardiotoxicity. National IDeA Symposium of Biomedical Research Excellence, July 2006, Washington, DC
88. Liang Q, **Kobayashi S**, Huang Y and Lackey T. Proteasomal Inhibition Preserves GATA4 Protein Levels and Attenuates Doxorubicin-induced Cardiomyocyte Death. Scientific Sessions 2005, American Heart Association, Dallas, TX.
89. **Kobayashi S**, Lackey T, Huang Y, and Liang Q. Transcription Factor GATA4 Directly Regulates Cardiac Bcl2 Gene Expression *in vitro* and *in vivo*. Scientific Sessions 2005, American Heart Association, Dallas, TX
90. **Kobayashi S**, Huang Y, Bisping E, Pu W, Molkentin JD, Liang Q. Bcl2 is a Preferred Gene Target for the Transcription Factor GATA4. 2nd Annual Symposium of the American Heart Association Council on Basic Cardiovascular Sciences, July 2005, Keystone, CO

91. Kuzman JA, Vogelsang KA, Liang Q, **Kobayashi S**, Thomas TA, and Gerdes AM (2005). Thyroid hormone activates AKT signaling in the adult rat heart and neonatal cardiomyocytes. 27th Annual ISHR American Section Meeting, 2005 New Orleans, LA

Grants and funding (list in descending chronological order, and include the granting institution, years funded, title, and dollar amount)

1. NIH Research Enhancement Award (1R15HL179642-01A1). Redefining The Role Of Lysosomal Membrane Permeabilization (LMP) In Doxorubicin-Induced Cardiotoxicity (PI), 07/15/2026 – 6/30/2029, \$573,750
2. American Osteopathic Association Research Grant. Fluid dynamics of concussion in mixed martial arts in a pilot randomized controlled trial of OMM, (co-PI, PI: Jayme Mancini, DO, PhD) 9/1/24-2/28/26, \$100,000
3. NIH Research Enhancement Award (1R15HL161737-01). Deciphering the role of lysosomal membrane permeabilization in Diabetic Cardiac Injury, (PI), 02/15/2022 – 1/31/2025, \$428,400
4. American Heart Association Scientist Development Grant (15SDG25080077). Lysosomal Dysfunction in the Diabetic Heart. Principal Investigator (PI), 7/1/2015 – 6/30/2019 \$308,000.
5. Juvenile Diabetes Research Foundation International Advanced Postdoctoral Fellowship (10-2008-457). Dissecting the Role of Autophagy in Type I Diabetes-Induced Cardiomyopathy. PI, 9/1/2008 – 8/31/2011 \$270,000
6. The American Heart Association Postdoctoral Fellowship (0625755Z). PI. Transcription Factor GATA4 Protects from Doxorubicin-induced Cardiotoxicity through Anti-apoptotic Gene Bcl2. PI, 7/1/2006 - 6/30/2008 \$100,476

Teaching experience (list in descending chronological order)

Apr. 1 2026-	(Course Director) MBSC 932 Research Methods in Biomedical Sciences
Sep. 1 2025-	Case-Based Learning (CBL)-1&2
Sep. 1 2022-	MBSC 933 Scientific Reading and Writing I, MBSC 934 Scientific Reading and Writing II, MBSC 935 Form and Function From Genotype to Phenotype I, MBSC 936 Form and Function From Genotype to Phenotype II, DO/PhD Program
Sep. 1 2021-	MBSC 930 Research Ethics, DO/PhD Program
Sep. 1 2020-	Research Methods in Biomedical Sciences, DO/PhD Program

Jan. 1 2018-	Academic Scholar Advanced Research Course
Oct.1 2013-present	Research Lab training for medical students and academic scholars in Department of Biomedical Sciences, NYCOM, Old Westbury, NY
Oct.1 1999-Mar. 31 2002	Molecular Biology Lab and Biology Lab for undergraduate students Department of Agriculture, Iwate University, Japan

Awards and honors (in descending chronological order).

1. Nominated as one of four finalists of the Roberto Bolli Competition for Young Faculty in Translational Science. "Lysosomal Membrane Permeabilization and The Ensuing Ectopic Expression of Cathepsin D Mediates Hyperglycemia-Induced Cardiomyocyte Death" at the 5th Annual Meetings of the International Academy of Cardiovascular Sciences (IACS): North American Section, on August 31 to September 2nd, 2017, Orland, FL.
2. Outstanding Research Award: "Diabetic Ketoacidosis Compromises Cardiac Mitochondrial Quality Control in Humans." **Kobayashi S**, Gilliland M.G.F, Hoffman W.H, O'Connor A, Jung MK, and Liang Q. The Second Annual ACRE/APS Symposium: Hot Topics in Current Basic and Translational Research of CV disease, July 12, 2015, New Orleans, LA.
3. **Kobayashi S**, Volden P, Timm D, Jepperson T and Liang Q. GATA4 Antagonizes Doxorubicin-induced Myocardial Injury by Suppressing Autophagy. The American Heart Association, Basic Cardiovascular Sciences 2010 Scientific Sessions, July 2010, Rancho Mirage, CA. (The abstract was selected as in the Best of AHA Specialty Conferences at Scientific Sessions. AHA Scientific Session, November 2010, Chicago, IL)
4. New Investigator Travel Award "Regulation of Akt/PKB activity by P21-activated Kinase in Cardiomyocytes", 4th Annual Symposium of the American Heart Association Council on Basic Cardiovascular Sciences, July 2007, Keystone, CO

Students mentored and in what capacity during last three years (may include NYCOM/NYIT students and other students)

Mentoring (* Research recognition award):

NYIT College of Osteopathic Medicine, 2013-Present:

Medical student researchers currently supervised:

2026 Johnathan Yarbrow (PhD Committee), Jack Chin, Zeeshan Khan, Raiya Patel, Irina Kapitanova, Aishwarya Selvaraman, Ronald Lospenuso, Sonia Patel, Giovanni Monaco, Sarah Chan, Martin Merizalde, Ramya Nadendla, Noa Darmon Asiag, Mayil Bhat,

Medical student researchers previously supervised:

2025 Jonathan Spadafora, Kieran Mannion, Gursimran Singh, Ronald Lospenuso, Rini Lee, Connor McGinley, Sonia Patel, Rohan Francis, Raymond Wong, Isabella Church, Ani

Rakoubian, Robert Grasso, Pravin Vathappallil, Zaen Asghar, Chris Mathew, Demarcus Nasef, Daniel Nasef, Christopher Lau, Alessia Vata, Pritpaul Singh, Brett Silverstein, **2024** Anoushka Guha, Pooja Jaiswal, Kinza Ahmed, **2023** Caleb Sooknanan, Lionel Chong, Anushka Bhatt, Harmeet Kaur, Teresa Zheng, Helen Deng **2022** Jennifer Van, Brett Silverstein, Hongling Chen, Steven Tseng, Naveed Tariq, Young Hee Hahn, Sameer Ahmad, Adeline Fleits, **2021** Tony Nguyen*, Akito Theodore Nicol, Zaid Tahir, Saahil Thukral, Ramim Shafi, Nabeel Sharaf, Lauren Soong, Jai Joshi, Thomas Jacob, Rejath Tose, Christopher Chen, Asad Sheikh, **2020** Mandeep Singh, Abdelrahman Abouelnas, Naunihal Singh, Vignesh Gunasekaran, Shawn Geffken, Zain Satti, Jane Gabriel, **2019** Jaclyn Del Pozzo*, Joy Patel*, Bharvi Marsha Patel, Sebastian Geraci, Almeet Kaur, Deep Patel, Moez Ahmed, Anjum Kazi, Anam Chaudhry, Bassel Hassan, George Economou, Asim Shahid, Muhammad Humanis, Katrina Bantis*, Salman Alam, Arjun Vidyasagar, Stacy Paickattu, Rutul Upadhyay, Omar Nagaria, Suprit Singh, Ryan Dunne, Romana Syed, Bethany Syracuse, Rizowana Kabir, Yawen Zhang, Kevin Clements, Jeremy Anthony Garcia, Amanda Kaminaris*, Tahsin Rahman, Rebecca Tamarkin, Usman Aslam*, Benjamin P Kramer*, Kathy Tin, Puja Mehta, Jesse Chait, Phillip Chin, Fiza Khan, Eric Lam, Michael Chang*, Max Cohen, Sophia Zigouras, Jedidah Nuamah, Angela Samaan, Mariam Atia, Ashley Weiner*, Chirag Parikh*, Neel Kachali, Valentyna Erstenyuk, Polina Pinkhasova, Lyudmila Burina, Michael Catanzaro, Sumir Shah, Nha Duong

NYIT undergraduate student researchers supervised:

Current:

Heba Ahmed, Chandrema Hossain

Previous:

Naman Kapoor, John Youssef, Alisha Marfani, Rubia Bindu, Mariam Makar, Chahat Arora, Jasky Kareem, Chahat Arora, Emily Joseph, Anamika Rao, Rodrigue Jean-Baptiste Jr., Anjani Bhavsar, Bhavsar, Pravin Vathappallil, Demarcus Nasef, Srinidhi Gadula, Nidhi Banker, Peter Girgis, Supriya Baskara, Tint Tha Ra Wun, Stephanie Cervini, Christina Cervini, Rahul Patel

High school student researchers supervised:

Sanford Research/University of South Dakota, 2006-2013

High School students: Troy Larsen

Volunteers: Mariko Hagiwara, Paul Volden

Student Outcomes (2006-Present):

Awards – National:

Tony Nguyen, AOA research training grant 2020-2021 Recipient (\$5,000)

Jaclyn Del Pozzo, Unexpected Role of Adenosine-Monophosphate Activated Protein Kinase in Doxorubicin Cardiotoxicity and Metformin-Mediated Cardioprotection. The 23rd Annual Scientific Meeting of Heart failure Society of America (HFSA), Philadelphia, PA. September 13-15, 2019. (Nominated as one of six finalists for oral presentation in the Jay N. Cohn New Investigator Award: Basic Science Session)

Katrina Bantis, AOA research training grant 2017-2018 Recipient (\$5,000)

Michael Chang, The Effect of Fasting on Mitochondrial Size and Mitophagy in Cardiomyocytes. Experimental Biology 2018 Apr 15-18 2018, San Diego, CA. (Nominated as a poster award finalist for an AAA Graduate Student Poster Award at AAA's Annual Meeting at EB 2018)

Katrina Bantis, AOA-OMED Conference 2017 Oct 7-Oct 10 2017, Philadelphia, PA. (Student Poster Competition 1st Prize Winner)

Amanda Kaminaris, ASBMB Graduate/Postdoctoral Travel Award—2017 Recipient (\$1,000), The American Society of Biochemistry and Molecular Biology, Experimental Biology 2017

Usman Aslam, Time-Dependent Differential Effects of Fasting on Cardiac Autophagy and Mitophagy. American Medical Association Foundation Seed Grant Research Program—2016 Recipient (\$2500),

Ashley Weiner, Mitophagy Contributes to Doxorubicin-Induced Cardiomyocyte Death. American College of Physicians, Internal Medicine Meeting 2016, May 5-7 2016, Washington D.C. (Young Achiever/Abstract Finalist in ACP's National Abstract Competition)

Parikh Chirag, The American Physiological Society, Experimental Biology 2016, April 2-6, 2016, San Diego, CA. (Excellent in Professional Student Research Travel Award—2016 Recipient (\$1800))

Usman Aslam, The American Physiological Society, Experimental Biology 2016, April 2-6, 2016, San Diego, CA. (Excellent in Professional Student Research Travel Award—2016 Recipient (\$1800), The American Physiological Society, Experimental Biology 2016)

Benjamin P Kramer, AOA-OMED Conference 2015 Oct 17-Oct 21 2015, Orland, FL. (Student Poster Competition 2nd Prize Winner).

Students pursuing a Higher or Professional degree:

Johnathan Yarbrow, DO/PhD, (2023-2028)

Tint Tha Ra Wun, BS student, UCONN MD/PhD (2026-2031)

Amanda Kaminaris, BS/PA student, NYIT SHP (2013-2017)

Tahsin Rahman, BS/DO student, NYIT COM (2013-2014)

Stacy Paickattu, BS/DO student, NYIT COM (2016)

Rutul Upadhyay, BS/DO student, NYIT COM (2016)

Usman Aslam, MS student in Clinical Epidemiology, Johns Hopkins Bloomberg School of Public Health (2015)

Other professional activities

Selected Oral Presentation at the American Physiology Summit 2026, "Live-cell imaging reveals altered mitochondria-lysosome contact dynamics in doxorubicin-induced cardiotoxicity." American Physiology Summit 2026, April 26, 2026, Minneapolis, MN

Invited talk at the Rutgers CBNP seminar series, titled "Organelle Crosstalk at the Heart of Cardio-Oncology: Focusing on Lysosomal Dysfunction," as part of their departmental academic program. Rutgers University, Newark, NJ Sep 29, 2025

Invited talk at the 61st CRC Seminar "Suppression of Autophagy-Lysosome Pathway Protects against Diabetic Heart Disease." Cryobiosystem Research Institute, Iwate University, Iwate, Japan Feb 13, 2012

Editorial Activities

Associate Editor: Molecular Medicine (Since 2022-)
Frontiers in Cardiovascular Medicine (Since 2023-)

***Ad hoc* Reviewer:**

AJP-Heart and Circulatory Physiology
Antioxidants
Archives of Biochemistry and Biophysics (ABB)
Autophagy
BMC Cardiac Research
Cardiovascular Research
Cell Communication and Signaling
Cells
Clinical Nutrition
Computational Biology and Chemistry
Endocrine and Metabolic Science
European Journal of Pharmacology
Frontiers in Cardiovascular Medicine
Frontiers in Pharmacology
Immunologic Research
International Immunopharmacology
International journal of Environmental Research and Public Health (IJERPH)
International Journal of Molecular Sciences (IJMS)
IUBMB life
Journal of Pharmacology and Experimental Therapeutics (JPET)
Journal of the American Heart Association (JAHA)
Medicina
Metabolites
Mitochondrion
Molecular Therapy - Nucleic Acids
Pharmaceutics
Scientific Reports

Community and Outreach service:

NYS Science & Engineering Fair Judge (2018, 2019, 2020)
LI Science & Engineering Fair Judge (2020-)

Other service activities (to NYCOM/NYIT, the osteopathic profession, and other)

Society Memberships (society name and year joined)

American Association for the Advancement of Science (AAAS) (2005)

American Heart Association (2007)

American Diabetes Association (2014)

The New York Academy of Sciences (NYAS) (2017)

American Physiology Society (2024)

Current research projects/activities

1. Molecular mechanisms of DOX-induced cardiotoxicity
2. Lysosomal dysfunction in the diabetic heart.
3. Organelle communications in cardiomyocytes

The projects are to elucidate the mechanisms that mediate diabetic cardiomyopathy and heart failure, a leading cause of death for diabetic patients.